

Models and Tools for Cyber-Physical Systems

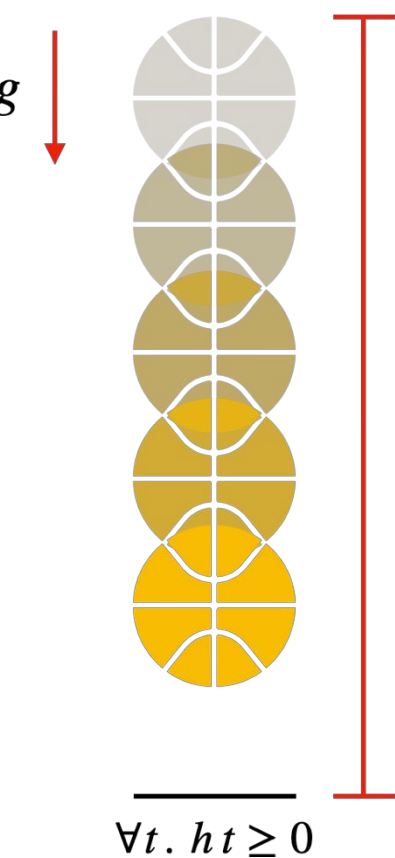
Chapter 9: Hybrid systems

Instructors: Suman Sourav, Jonathan Julián Huerta y Munive
{sumansourav,jjhymuni}@cs.aau.dk

Original slides courtesy of Rajeev Alur
alur@cis.upenn.edu

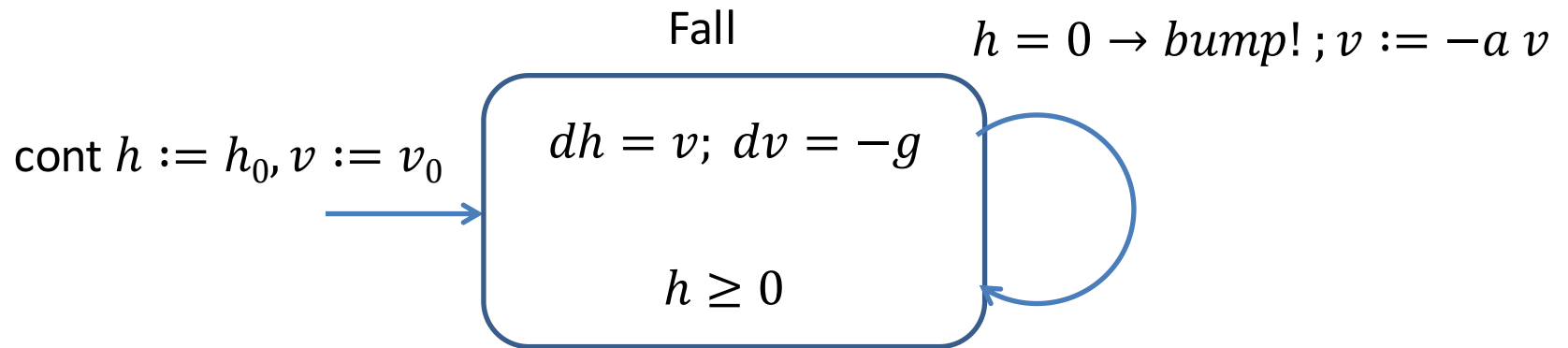
Zeno behavior

Modeling a Bouncing Ball



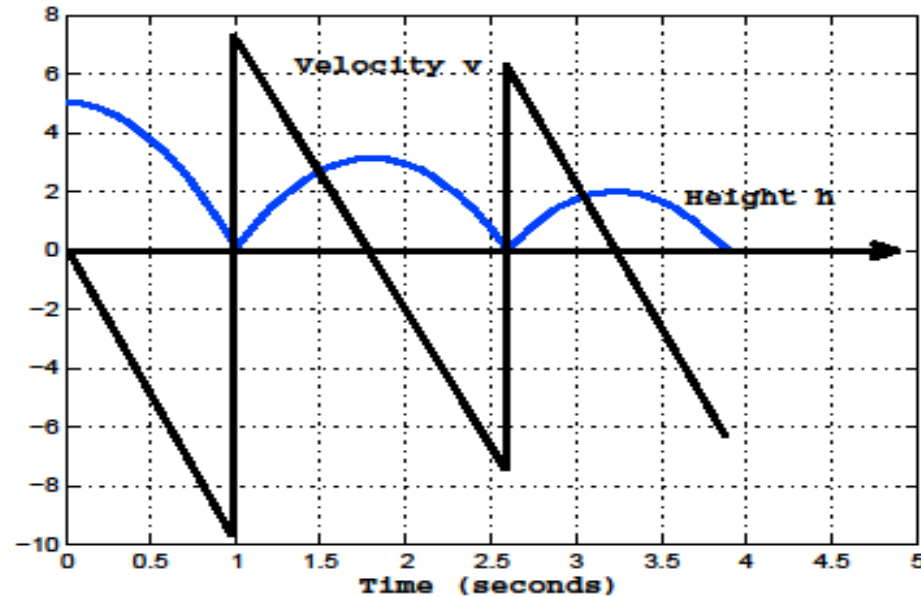
- ❑ Ball dropped from an initial height h_0 with an initial velocity v_0
- ❑ Velocity changes according to the differential equation $dv/dt = -g$
- ❑ When the ball hits the ground, that is, when height $h = 0$, velocity changes discretely: $v := -a v$, where $0 < a < 1$ is damping constant
- ❑ Modeled as a hybrid system: mix of discrete and continuous behaviors!

Hybrid Process for Bouncing Ball

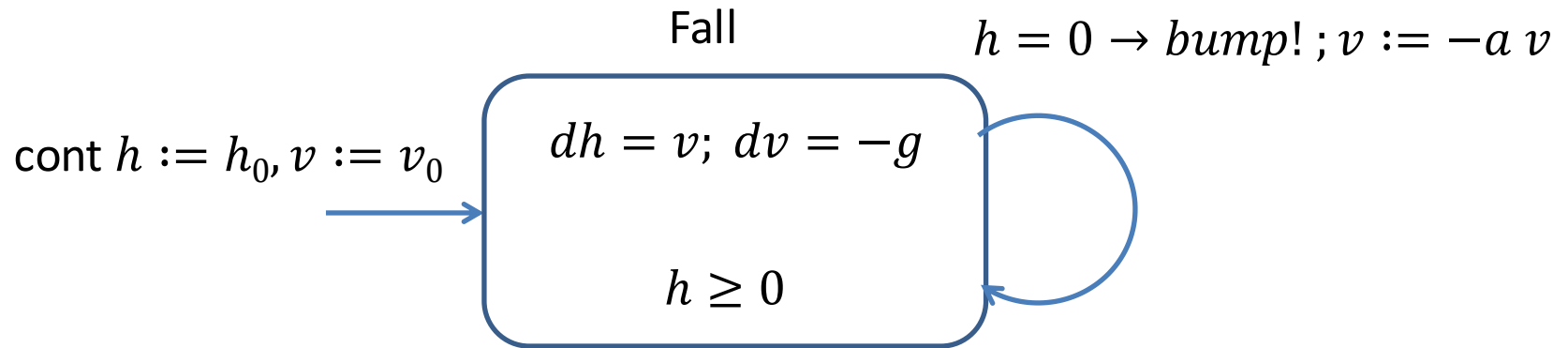


Execution of the BouncingBall process

$$h_0 = 5; v_0 = 0$$



Analysis of Bouncing Ball Model



Change in height during first bounce: $\mathbf{h}(t) = h_0 - gt^2/2$

Time at which first bump occurs: $t_1 = \sqrt{2h_0/g}$

Velocity just before first bump occurs: $-\sqrt{2gh_0} = -v_1$

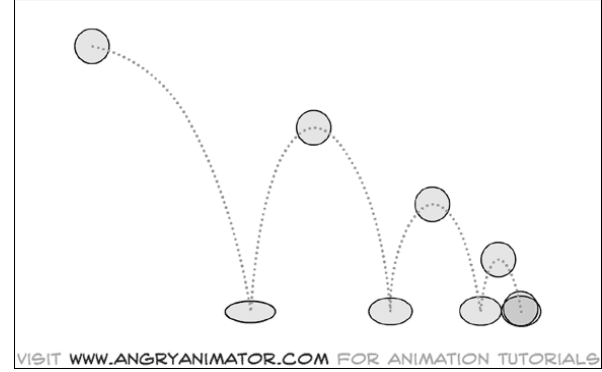
Velocity just after first bump : $v_2 = av_1$

Evolution of height during second bounce: $\mathbf{h}(t) = v_2t - gt^2/2$

Time between first and second bump: $t_2 = 2v_2/g$

Velocity just before second bump occurs: $-v_2$ and after second bump $v_3 = av_2$

Modeling a Bouncing Ball

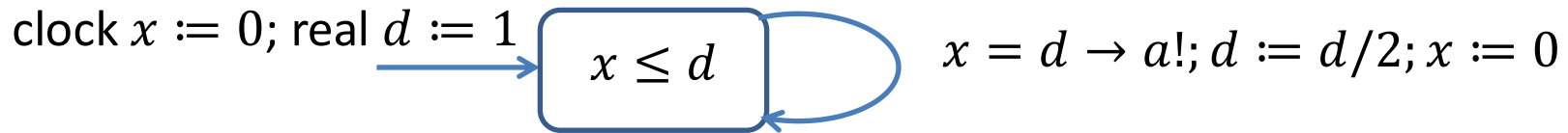


- ❑ Velocity after k bumps = $a^k v_1$
- ❑ Duration between k -th and following bump $a^k v_1 / g$
- ❑ Sum of all durations converges to some finite K
- ❑ Infinitely many discrete actions in finite time = Zeno behavior!
- ❑ An execution with infinitely many discrete actions describes behavior only up to time a certain time K and does not tell us the state of the system beyond K

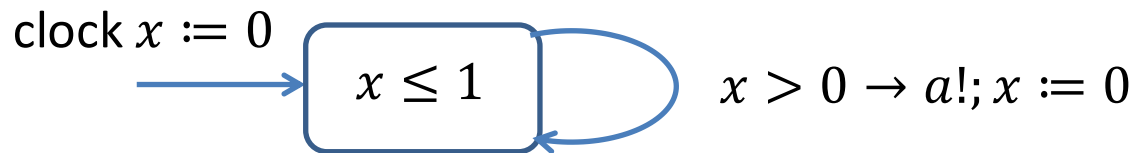
Formalization

- ❑ An infinite execution of a hybrid process HP is of the form $s_0 \xrightarrow{t_1} s_1 \xrightarrow{t_2} s_2 \xrightarrow{t_3} s_3 \dots$, where t_i is duration of i -th step
 - Input/output/internal actions are instantaneous (duration 0)
- ❑ An infinite execution is called Zeno if the infinite sum of all the durations is bounded by a constant, and non-Zeno if the sum diverges
- ❑ A state s of the process HP is called
 - Zeno if every execution starting in state s is Zeno
 - Non-Zeno if there exists some non-Zeno execution starting in s
- ❑ A hybrid process HP is called non-Zeno if every reachable state of HP is non-Zeno
 - At every point during an execution it is possible for time to diverge
- ❑ Zeno system: Could end up in a state from which duration between successive steps must get smaller and smaller
- ❑ Thermostat: non-Zeno; Bouncing Ball: Zeno

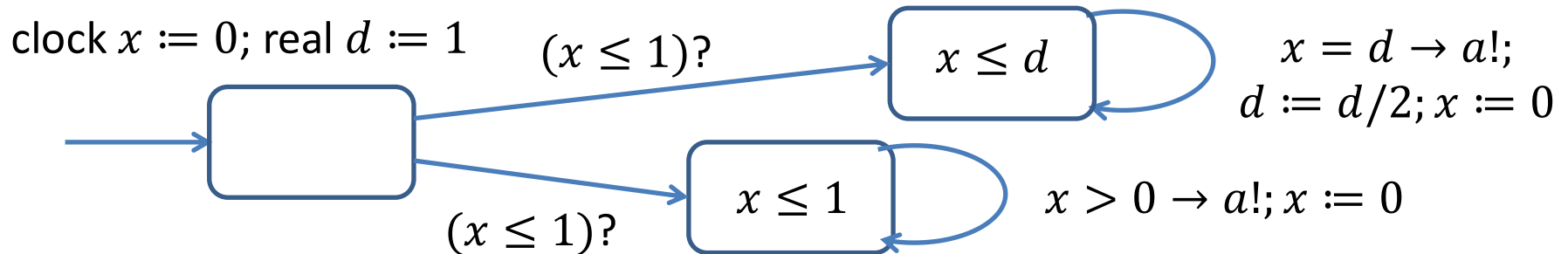
Zeno Vs Non-Zeno



Zeno! Every possible execution is Zeno



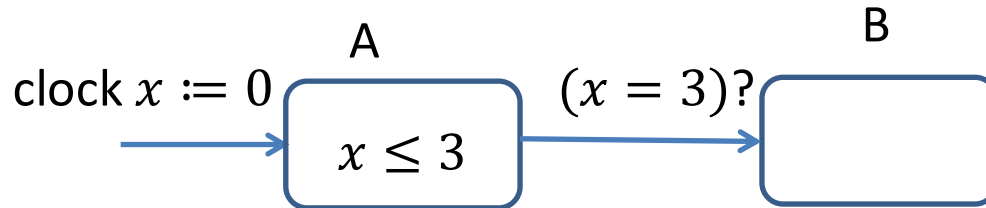
Non-Zeno! Some executions are Zeno and some are non-Zeno



Zeno! System may end up in a state from which only Zeno executions are possible

Zeno Processes and Reachability

- ❑ How does existence of Zeno processes influence analysis?
- ❑ Recall: A state s is said to be reachable if there exists a finite execution starting in an initial state and ending in state s
- ❑ Safety: A property φ is an invariant if all reachable states satisfy φ

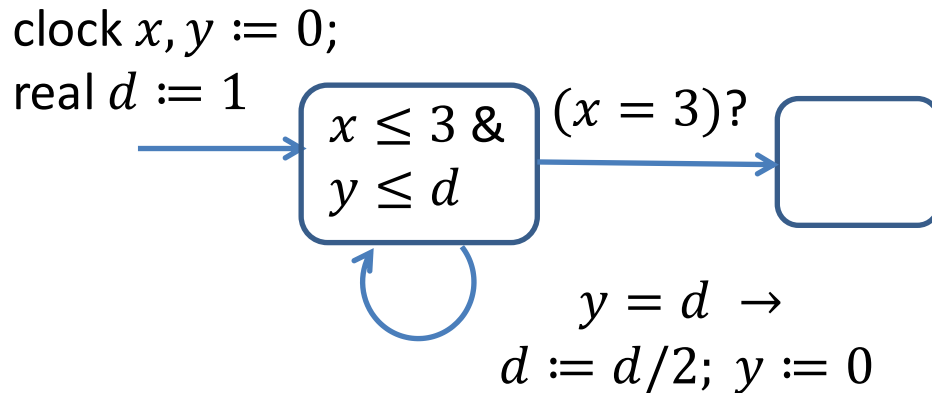


Is mode B reachable ?

Zeno Processes and Reachability

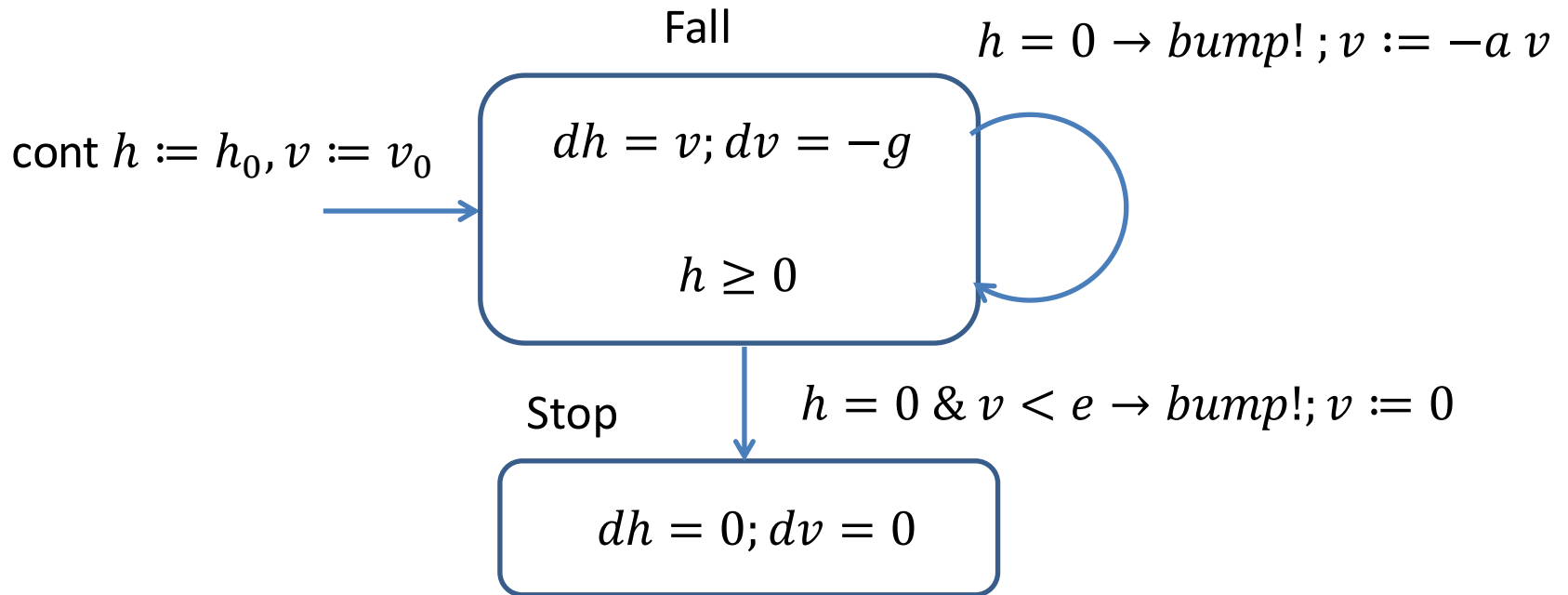


Is mode B reachable ?



Presence of a Zeno process in the system can stop time from advancing and make states of other processes unreachable !

Making Bouncing Ball Non-Zeno



If velocity is too small, stop modeling dynamics accurately

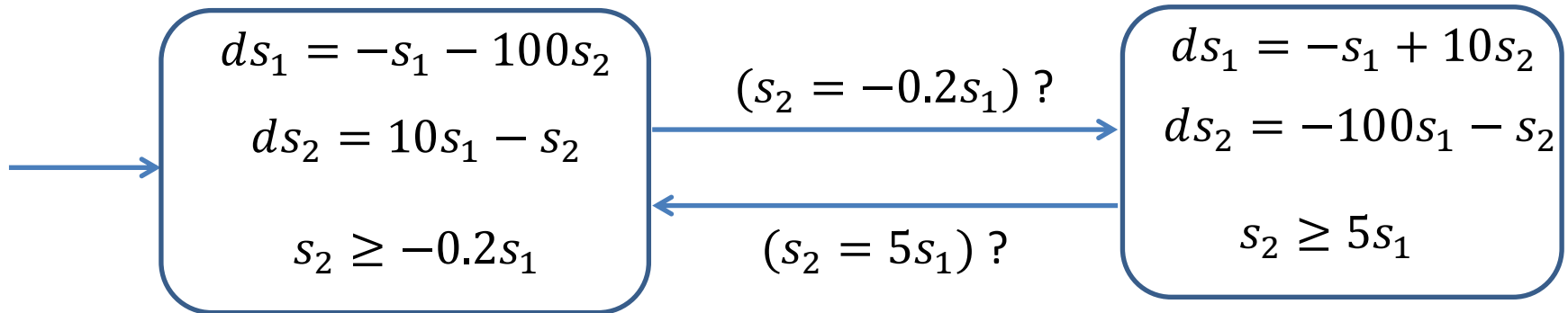
In this model, there is a lower bound on duration between successive bumps

Stability analysis

Stability of Hybrid Systems

A

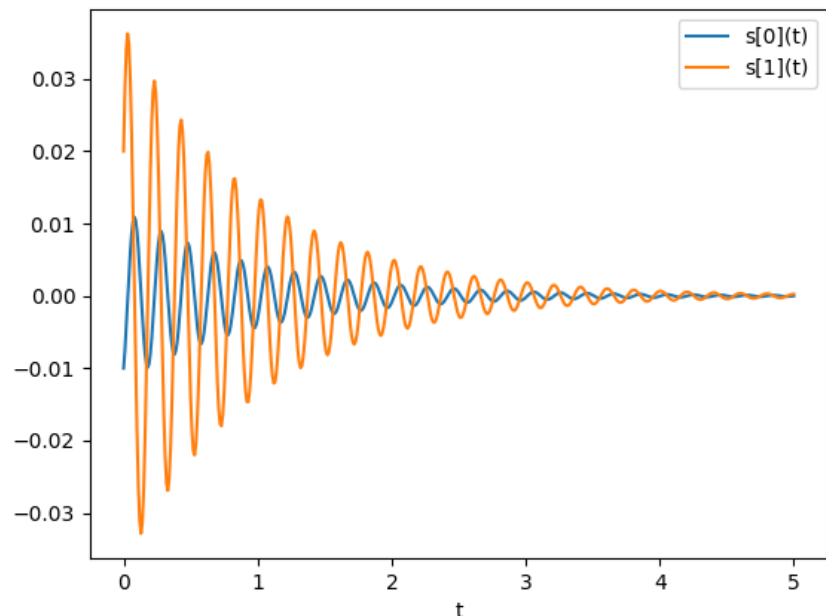
B



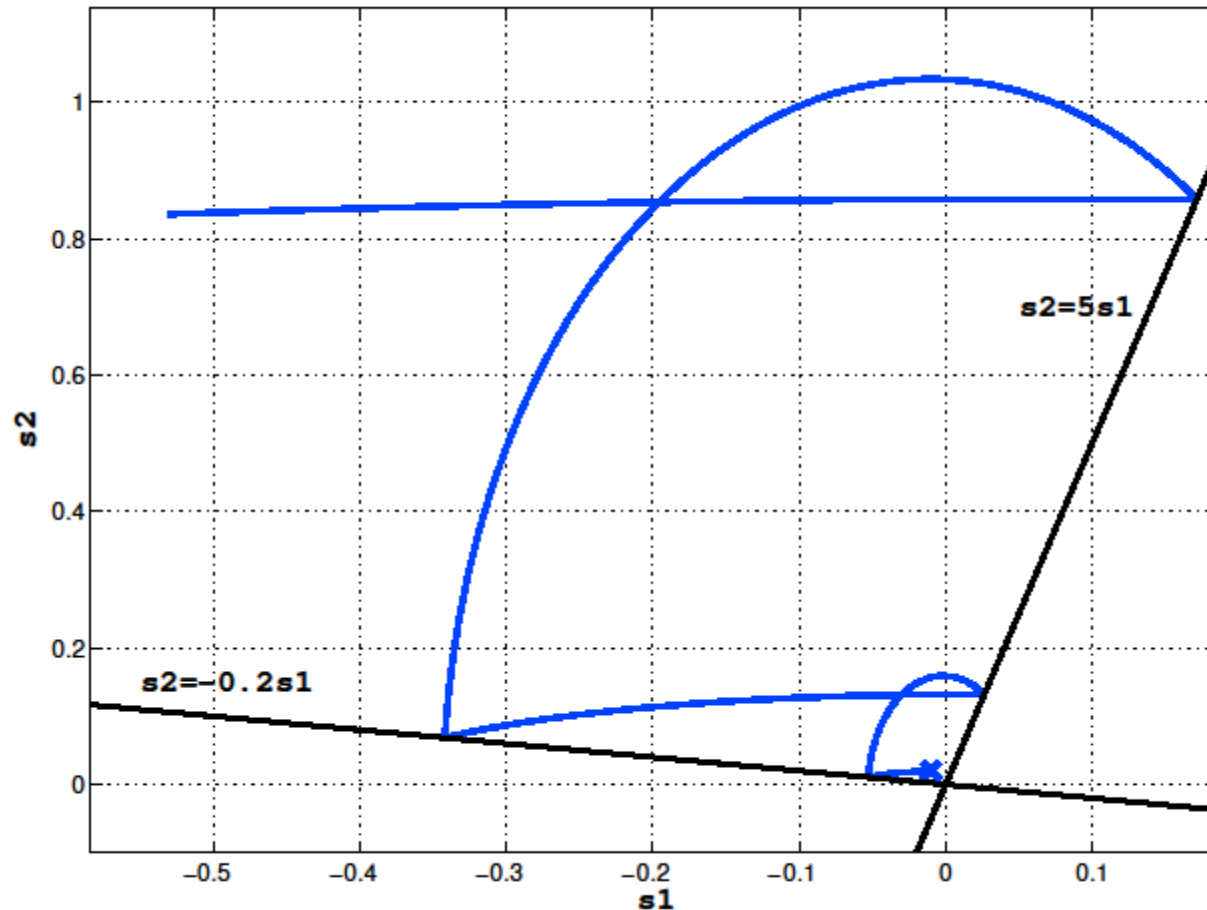
Is the dynamics in mode A stable?

Is the dynamics in mode B stable?

Both modes have stable dynamics, k



Stability of Hybrid Systems



Stabilizing unstable systems

Equilibrium state 1: $v = 0$; $\varphi = 0$;
Pendulum is vertically downwards.

Stable, but not asymptotically stable

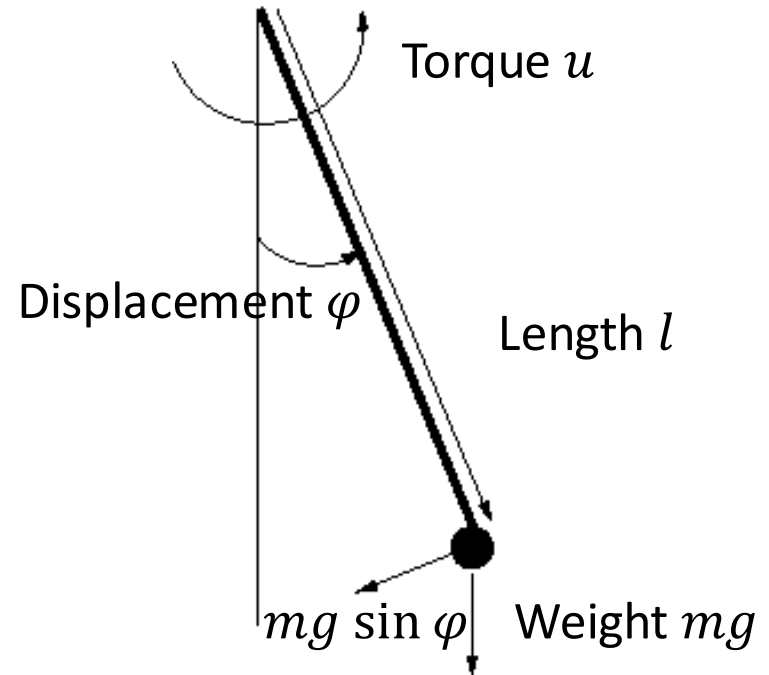
Equilibrium state 2: $v = 0$; $\varphi = -\pi$;
Pendulum is vertically upwards

Unstable !

But a hybrid controller can solve this problem!

<https://youtu.be/5oGYCxxkgnHQ>

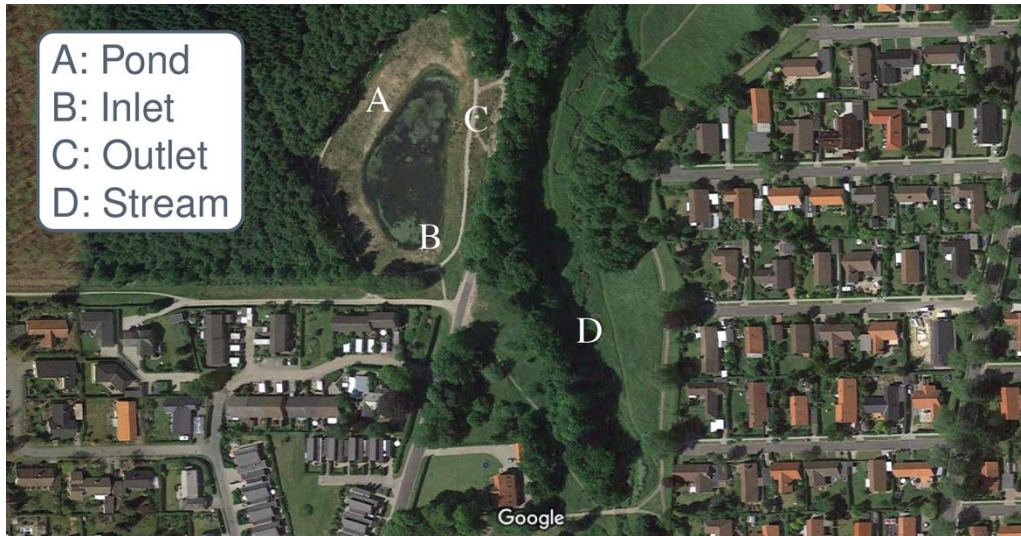
<https://www.hep.phy.cam.ac.uk/~lester/teaching/PartIIIProjects/2017-PeterDolan-InvertedPendulum.pdf>



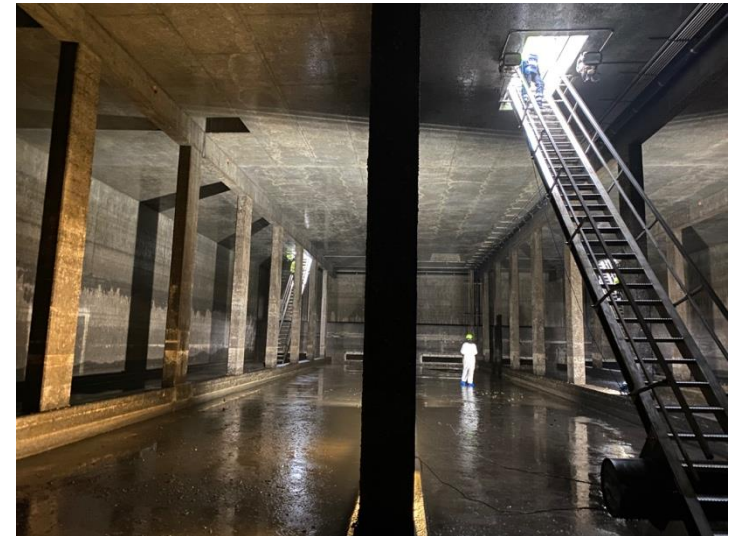
Research example

Storm water detention ponds

Vilhelmsborg Skov pond, Aarhus



Grønjordsvej basin, Copenhagen

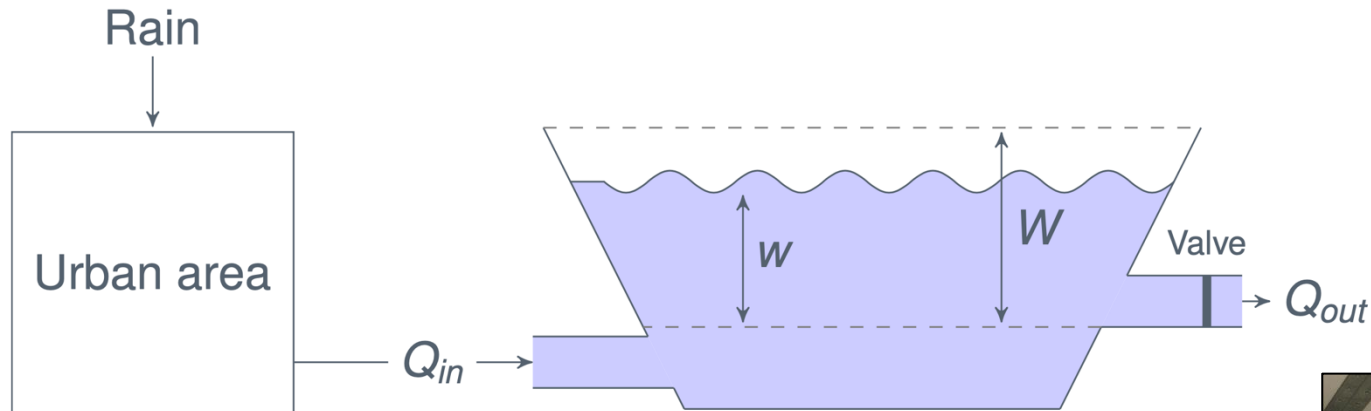


Ponds are designed to reduce:

- Urban area flooding
- Downstream flooding
- Stream bed erosion
- Natural disturbances

Controlling stormwater detention ponds under partial observability.
Journal of Logical and Algebraic
Methods in Programming
<https://www.sciencedirect.com/science/article/pii/S2352220824000336>

Schematic overview



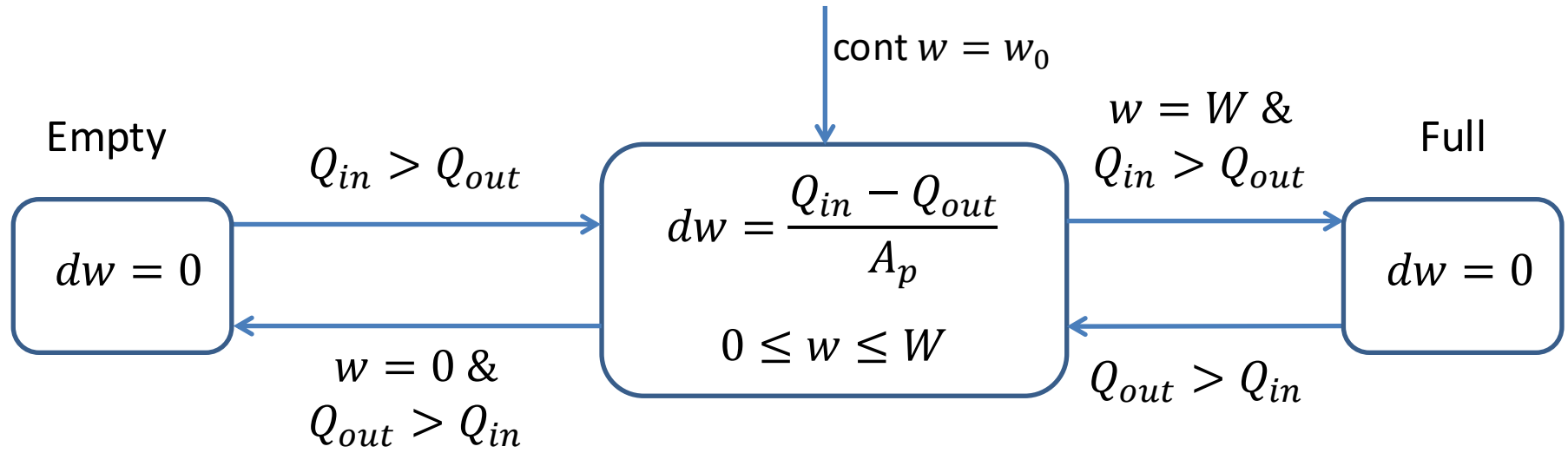
- w : water level
- W : max water level
- Q_{in} : input flow
- Q_{out} : output flow

The model specification includes:

- Pond geometry
- Controller (valve)
- Rain
- Urban area



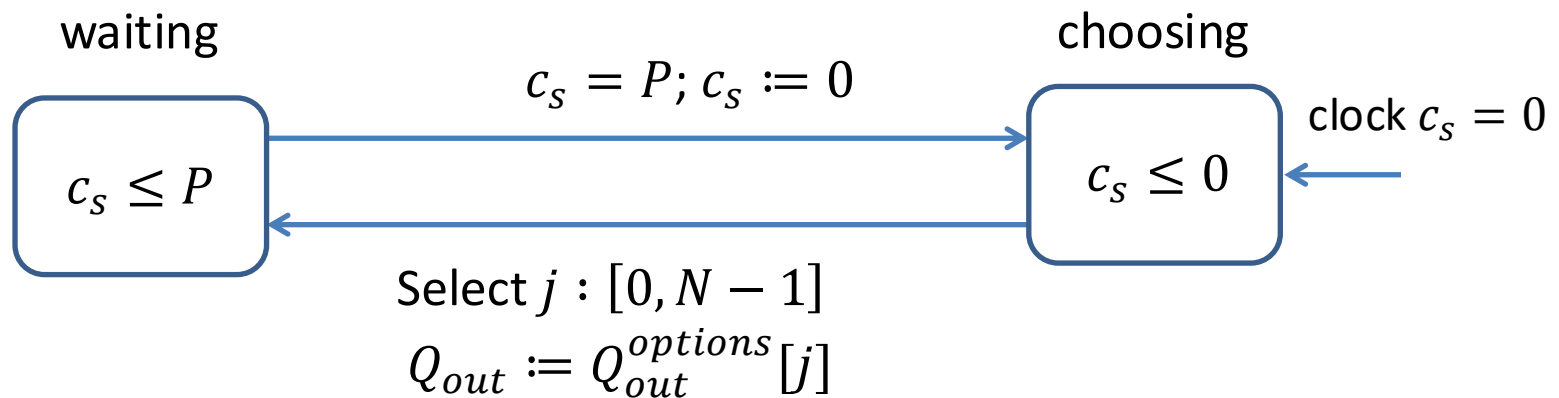
Hybrid model of the pond



The model of the controller

The model of the controller is a timed automata:

- ❑ The controller can only change the setting of the pond's valve periodically P (due to energy constraints)
- ❑ The controller can change the setting only to a limited number N of pre-defined settings



Controller requirements

Strategy

- ☐ A strategy determines which control mode for the output flow Q_{out} can be used in the next control period.

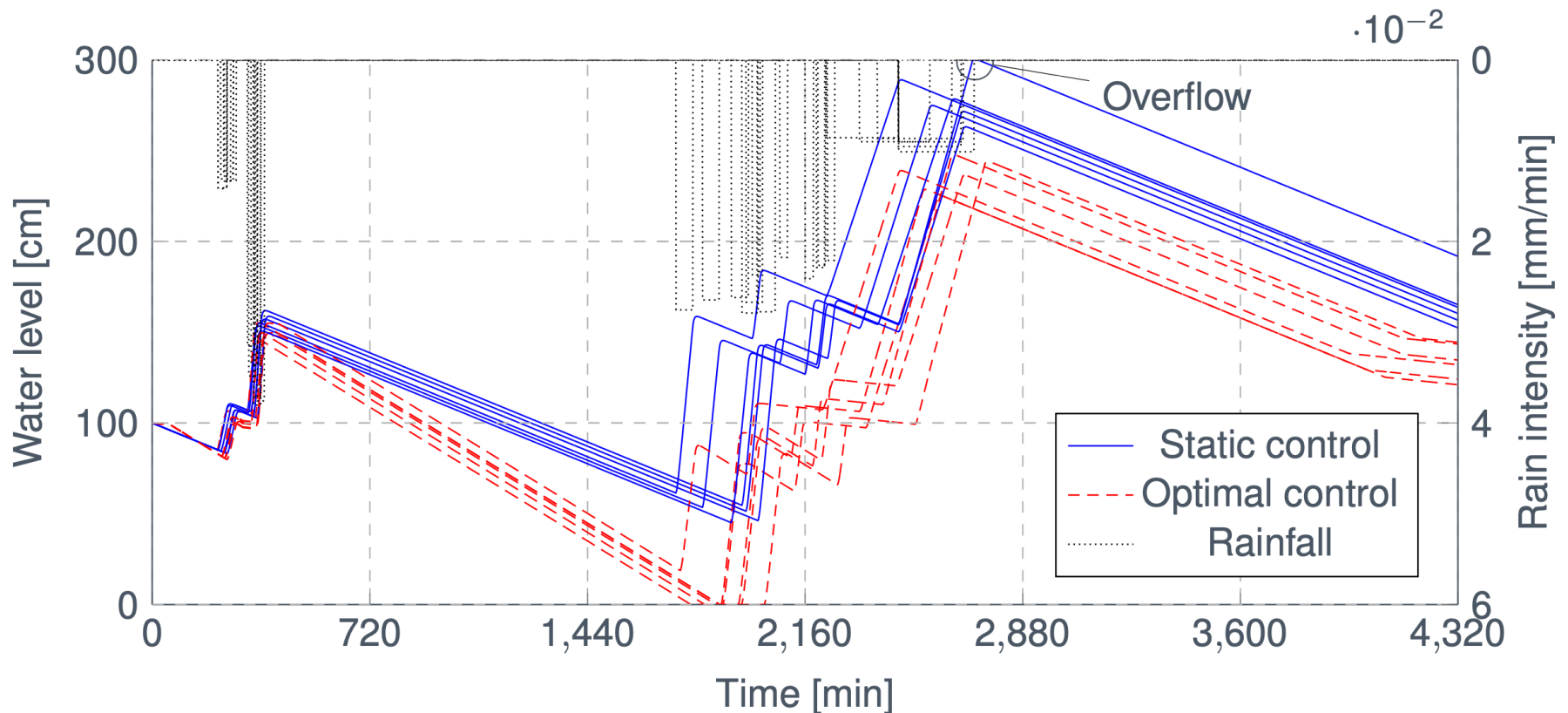
Safe strategy

- ☐ A strategy is safe with respect to a set of safe states if all states encountered are within the safe set.
- ☐ For the pond, a safe strategy does not let the pond overflow.

Optimal strategy

- ☐ A strategy is optimal if it minimizes or maximizes the expected value of an optimization variable.
- ☐ For the pond, we maximize the particle sedimentation by maximizing the water level.

Static control versus safe and optimal control



Ongoing and future work

Ensure robust pond control under erroneous and noisy sensors

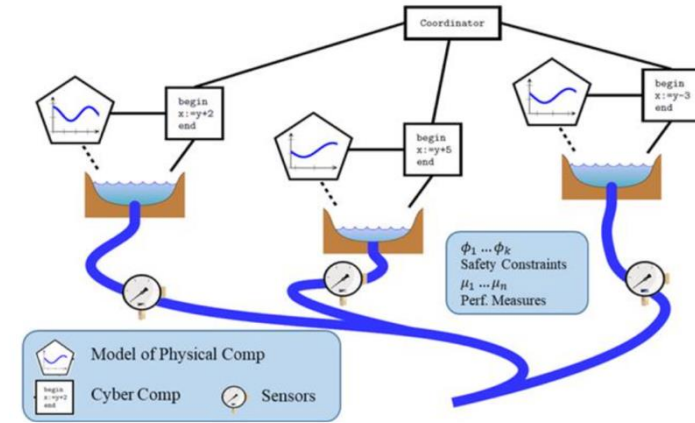
- ❑ Explicitly incorporate other data sources and noise models in the control model

Scalable and decentralized control synthesis for multiple ponds

- ❑ Distributed control synthesis

Machine learning models of soft sensors:

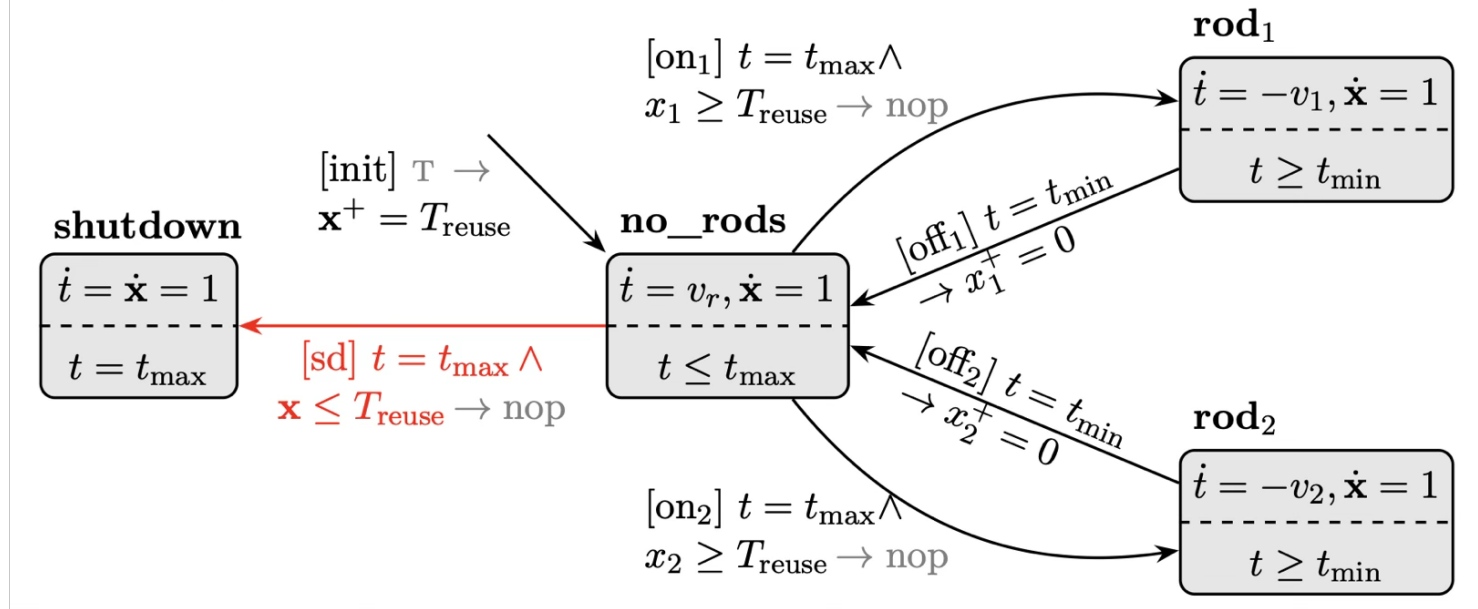
- ❑ Water height/flow estimation based on indirect sensor measurements



Challenge exercise

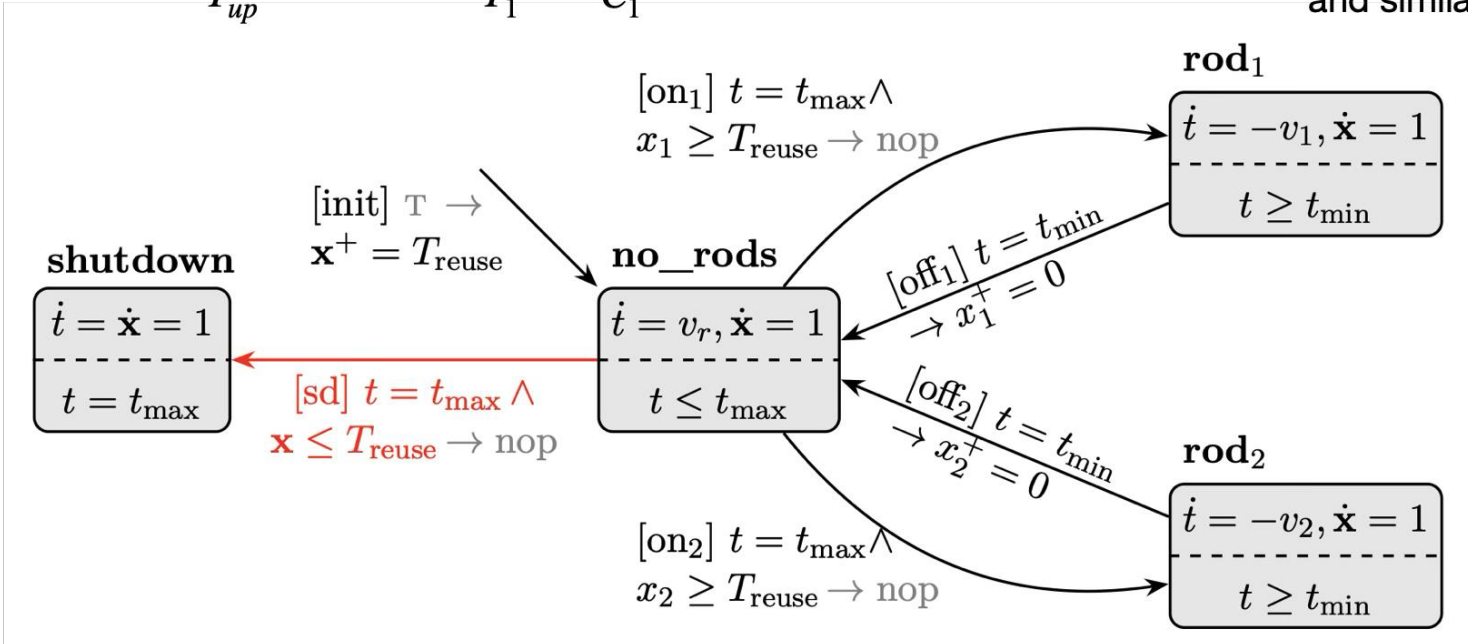
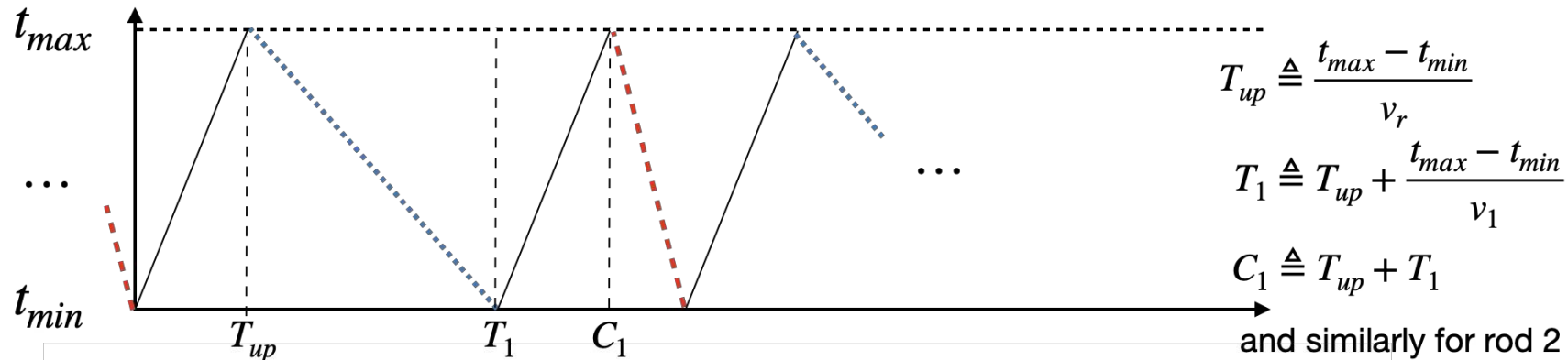
Consider a reactor with two rods (called rod_1 and rod_2) that can be applied (one rod at a time) to moderate the reaction and decrease the temperature t of the coolant, whose temperature has to stay within $[t_{\min}, t_{\max}] = [250, 1000]^\circ\text{C}$. The use of rod_1 and rod_2 results in the absolute temperature-decrease rates of $v_1 = 25^\circ\text{C}$ respectively $v_2 = 10^\circ\text{C}$ per time unit. The unmoderated reactor exhibits a temperature-increase rate of $v_r = 34^\circ\text{C}$ per time unit. Each of the rods can only be reused after a pause of at least $T_{\text{reuse}} = 80$ time units. If the reactor reaches the upper bound $t_{\max}$ before any of the two rods can be reused then it is shut down.

Show that it never reaches the shutdown mode



Hint: A graph of the behaviour of the temperature over time might help you propose an invariant conjoining 3 cases: when the rod 1 is applied, when the rod 2 is applied, and when no rods are applied. You might have to add stopwatches to your control.

Challenge exercise



Hint: A graph of the behaviour of the temperature over time might help you propose an invariant conjoining 3 cases: when the rod 1 is applied, when the rod 2 is applied, and when no rods are applied. You might have to add stopwatches to your control.